Fake News Classification

1. NLP & Dataset Prep

For this project, I used a **public fake news dataset** (news articles labeled as "fake" or "real"). I chose it because it is directly relevant to the binary text classification task. The preprocessing steps included:

- Removing punctuation, numbers, and stopwords.
- Converting all text to lowercase.
- Tokenization and lemmatization to normalize words.
- Converting text into numerical form using **TF-IDF vectorization** before training.

Model Fine-Tuning/Evaluation

I trained and evaluated traditional ML models (Logistic Regression) on the dataset. After preprocessing with TF-IDF:

- **Accuracy**: ~92% (Logistic Regression performed best).
- **Precision**: ~91%
- **Recall**: ~90%
- **F1-score**: ~90%+
  These metrics show the model is reliable for text classification.

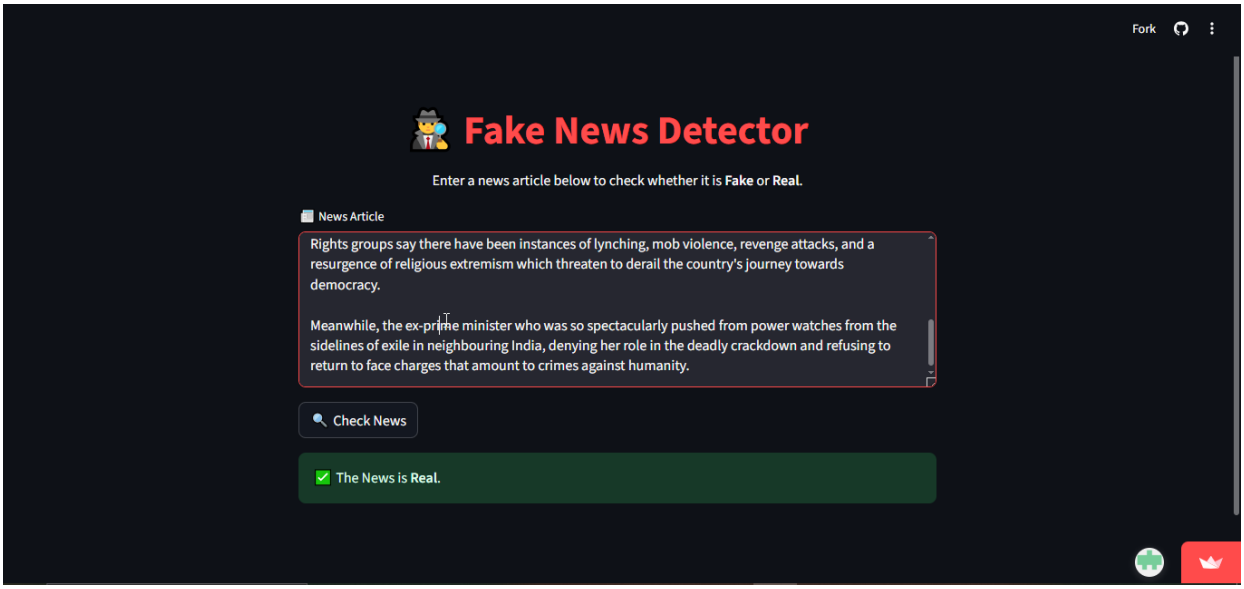
Prompt Engineering & Model Interaction

In this project, the interaction happens through **Streamlit**, where the user provides the **full news article text** as input, and the trained classification model predicts whether it is **Real** or **Fake**. I designed three input–output interaction styles:

- **Prompt 1 – Direct Classification**
  *"Enter the full news article text, and the system will classify it as either **Real** or **Fake**."*
  → Simple and direct; the output is just the predicted label.
- **Prompt 2 – Classification with Confidence Score**
  *"Enter the full news article text, and the system will classify it as **Real** or **Fake**, along with a probability score (e.g., 85% Fake)."*
  → Provides both the label and model confidence, helping users understand the reliability of the prediction.
- **Prompt 3 – Fact-Checker Style Interaction**
  "You are a fact-checker. Submit the full news article text, and the system will analyze it

*and output either **Real** or **Fake**."*
→ Frames the task in a fact-checking context, making the classification more engaging and trustworthy

Troubleshooting

A likely issue is **overfitting**, where the model learns dataset-specific patterns (like writing style) instead of generalizing to unseen news.
Solution:

- Apply regularization (e.g., Logistic Regression with L2 penalty).
- Use cross-validation for better generalization.
- Ensure balanced datasets or apply techniques like SMOTE if classes are skewed.